Green MIMO Detection

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1. ABSTRACT

In this paper, a low-complexity K-Best detector is presented for MIMO systems. To balance performance and complexity, the number of candidate nodes retained at each layer is fixed at K. A sorting mechanism is used to enhance the efficiency of the K-Best search [1]. Simulation and complexity results show that competitive performance can be achieved with modest complexity.

2. Proposed Green MIMO Detection

Consider an $N_r \times N_t$ MIMO system, where N_r and N_t are the numbers of receive and transmit antennas. Then, the real-valued system model can be expressed as $\boldsymbol{y_r} = \boldsymbol{H_r}\boldsymbol{x_r} + \boldsymbol{n_r}$, where $\boldsymbol{y_r}$, $\boldsymbol{H_r}$, $\boldsymbol{x_r}$, $\boldsymbol{n_r}$ are the received vector, fading channel matrix, transmitted M-QAM symbol vector, complex AWGN vector, respectively.

The proposed detector consists of three stages: 1) Sorting the channel column vectors to rank the detection order; 2) Conducting QR-decomposition on $\boldsymbol{H_r}$ to find the corresponding the unitary matrix $\boldsymbol{Q}$ and the upper triangular matrix $\boldsymbol{R}$; and 3) Carrying out K-Best mechanism to find the estimated symbols, expressed as $\widehat{\boldsymbol{x}_r}_{MLD} = \arg min \|\tilde{\boldsymbol{y}} - \boldsymbol{R}\boldsymbol{x_r}\|^2$, where $\tilde{\boldsymbol{y}} = \boldsymbol{Q}^H \boldsymbol{y_r}$. In fact, the K-Best mechanism computes the partial Euclidean distance (PED) at each layer. The *PED* at the i-th layer can be described as

$$PED_i = PED_{i+1} + \left|\tilde{y}_i - \sum_{j=i}^{2N_t} r_{i,j} x_j\right|^2,\ PED_{2N_t+1} = 0 \tag{1}$$

References

[1] Y.-X. Liu, et al, "An efficient K-Best MIMO detector for large modulation constellations," *IEEE Open Journal of Circuits and Systems*, vol. 5, pp. 2-16, 2024.

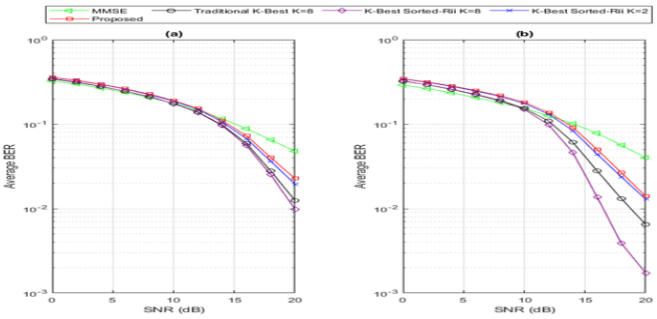

Fig. 1. BER comparisons with 64-QAM modulation, and (a) $N_t = N_r = 4$, and (b) $N_t = N_r = 8$.

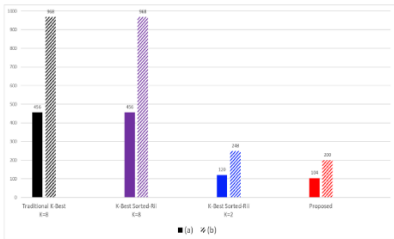

Fig. 2. The number of search nodes with 64-QAM modulation, and (a) $N_t = N_r = 4$, and (b) $N_t = N_r = 8$.